Research Statement

by Wassim Kabalan

I am a PhD candidate in computational cosmology at the Astroparticule et Cosmologie laboratory in Paris. My research spans *cosmic microwave background (CMB)* analysis and *large-scale structure (LSS)*, with a common focus on open-source, Python/**JAX** pipelines that are both differentiable and distributed for next-generation surveys such as `Euclid` and `LSST`. On the CMB side, I develop a spatially varying parametric component-separation method with the aim to *reduce bias in the tensor-to-scalar ratio* r for *Simons Observatory (SO)*–like data. I am also heavily involved in the open-source development of software packages for the cosmology community. I lead `jaxdecomp` (multi-GPU 3D domain decomposition and distributed FFTs) and I am one of the main contributors to `JaxPM` (differentiable particle–mesh), enabling gradient-based *weak-lensing inference* at *Rubin–LSST* scale. I prioritize open-source, reproducible, survey-ready software—from laptops to supercomputers—with end-to-end uncertainty propagation.

1 Summary of previous/current research

CMB: Spatially Varying Component Separation For *Simons Observatory (SO)*, *primordial B-modes* test inflation via the *tensor-to-scalar ratio* r (Kamionkowski and Kovetz, 2016). Current limits are dominated by *spatially varying Galactic foregrounds*: enforcing sky-constant spectral parameters leaks power into large-scale B and *biases* r (Planck Collaboration, 2020a). Without proper modeling of the spatial variability of the foregrounds' *Spectral Energy Distributions (SEDs)*, component-separation methods will leave significant residuals in the cleaned CMB maps, leading to biased estimates of the tensor-to-scalar ratio r and potentially obscuring the primordial B-mode signal. Finding optimal spatial partitions is challenging and computationally expensive, as current parametric methods such as FGBuster (Poletti, 2023) are far too slow to scan the large configuration space of possible partitions.

To address this challenge, I developed a novel spatially varying parametric component-separation framework on the sphere within **FURAX** (Chanial et al., 2024a), a **JAX**-powered (Bradbury et al., 2018) library enabling fast component separation routines. The end-to-end differentiable pipeline leverages GPU acceleration to achieve approximately **200**× speedup over traditional CPU-based codes such as **FGBuster**, allowing for millions of component-separation runs in a reasonable time frame. This throughput lets us scan partitions that jointly minimize foreground residuals and variance; on LiteBIRD-like simulations our work lowers the 68% upper limit on r compared to a sky-constant baseline (Fig. 1) and produces a 44% improvement compared to state-of-the-art spatially varying component-separation methods.

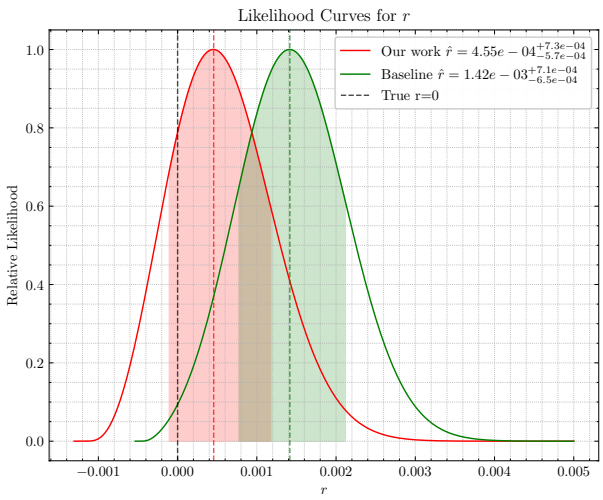

Figure 1: Posterior on r for SO-like skies. Spatially varying **FURAX** (orange) reduces foreground residuals and tightens the 68% upper-limit on r by **44%** relative to a sky-constant baseline (gray).

This work relies on sphere clustering routines I developed by contributing to `jax-healpy` (Chanial et al., 2024b), a **JAX**-native reimplementation of **HEALPix** (Górski et al., 2005) functionalities.

The pipeline is run using **jax-grid-search** on the **Jean Zay supercomputer**, a distributed hyperparameter scan library I developed to efficiently explore large parameter spaces across multiple GPUs by distributing the workload.

This framework is being integrated into the **MegaTop** pipeline for *SO* and will be a major component for future *CMB-S4* analyses.

LSS: Map-level Inference with Differentiable, Distributed Particle–Mesh *Weak gravitational lensing* encodes the projected matter distribution in shear maps, and thus it is an important probe for cosmological models that sheds light on the nature of dark matter and dark energy. At *LSST* depth and area, the field is strongly *non-Gaussian*, which renders traditional two-point statistics insufficient to capture all the information (Mandelbaum, 2018; Petri et al., 2016). A forward-model, map-level inference approach is therefore essential. Existing tools such as **PWMD** (Li and collaborators, 2024) and **JAXPM** (Lanusse et al., 2020) provide differentiable particle–mesh (PM) simulations, but they are limited to single-GPU and small volumes, so running an inference-capable pipeline from initial conditions to weak-lensing maps at *LSST* scale remains an open challenge.

To address this challenge, I created **jaxdecomp** (Kabalan and Lanusse, 2025), a **JAX**-native library for 3D domain decomposition and communication-intensive primitives (distributed FFTs, all-to-alls, halo exchanges) with performance-critical **CUDA** implementations. This package enables multi-GPU PM simulations and is integrated into the new version of **JaxPM** to run large-scale, differentiable cosmological simulations across multiple GPUs (Fig. 2).

I validated this framework by running cosmological simulations at a massive scale of 2048^3 particles with **JaxPM** on 64 *A100* GPUs on the **Jean Zay** supercomputer, achieving scalable performance and end-to-end differentiability. This capacity to run and sample large-scale, differentiable forward models is crucial for accurate weak-lensing analysis at *LSST* scale.

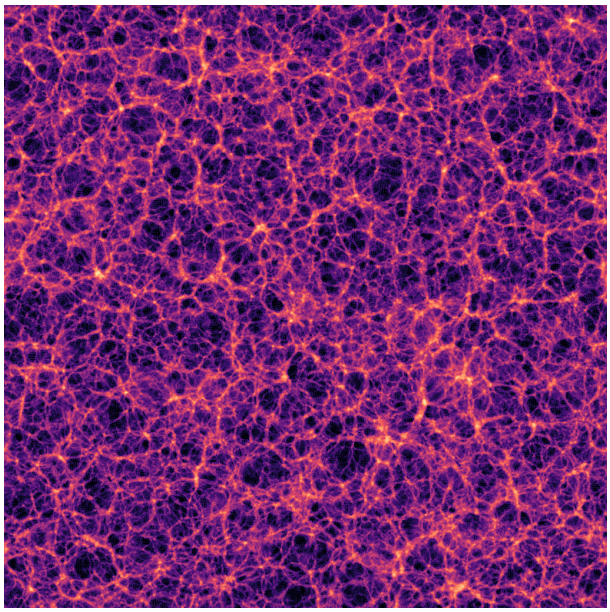

Figure 2: A *Lagrangian Perturbation Theory (LPT)* density field at $z = 0$ from a 2048^3-particle simulation with **JaxPM**, which I ran on 64 *A100* GPUs on **Jean Zay HPC**. A single step of LPT takes 2 seconds.

I am currently integrating the remaining steps required to model weak-lensing observations—light-cone assembly, spherical projection, and Born-lensing approximation—in a fully differentiable and distributed manner. As a first step towards full weak-lensing forward modeling, I am running a preliminary pipeline from initial conditions to spherical *convergence* (κ).

2 Future Work: Scientific Roadmap

The central motivation of my future research is the persistent σ_8/S_8 *tension*: late–time *weak-lensing* constraints remain lower than early–time *CMB* in recent surveys (e.g., *KiDS-1000*, *DES Y6*, *Planck 2018*) (Heymans et al., 2021; Jeffrey et al., 2024; Planck Collaboration, 2020b). My goal is to develop and apply differentiable, field-level inference that propagates *structure growth*, *gravitational lensing*, and *survey systematics* (including *intrinsic alignments*) to the map level, explicitly capturing *non-Gaussian* information while remaining interpretable in the physical field and extensible to *baryonic* physics.

These so-called *explicit inference* approaches are promising methods to fully leverage the information content of upcoming surveys such as *Rubin–LSST* and *Euclid* due to their interpretability, extensibility, and ability to capture complex non-linear information.

My research roadmap has several main milestones:

Before graduating from my PhD, I will deliver the first field-level constraints on *DES Y6* maps using a validated forward model from Gaussian initial conditions to spherical *shear* maps. My scientific goal is to quantify the gain from *non-Gaussian* information and establish bias control for S_8. The pipeline validates a Born-lensing path against compact ray-tracing benchmarks and produces shear maps using **s2fft** (Price et al., 2024), to which I contribute. A first step is applying this pipeline to *DES* as a precursor to *LSST* DR1 analysis.

Mid-term, I will quantify and control *baryonic* systematics at *LSST* scales. I aim to implement a physics-informed, differentiable *baryonification* inside **JaxPM**, operating on light-cone shells with interpretable parameters tied to *gas thermodynamics* and *feedback*, calibrated to hydrodynamical targets and statistics beyond the *power spectrum* (Schneider et al., 2025). These modifications propagate through the *weak-lensing* forward model while preserving large-scale modes and capturing small-scale suppression. Joint information from *CMB lensing* and *tSZ*/*kSZ*—where *LSST* overlaps *SO*/*CMB-S4*—acts to break the *baryon–cosmology* degeneracy and stabilize small-scale inference.

The long-term goal is to meet *Euclid* accuracy across area and angular scales. This requires a forward model with sufficient resolution to capture small-scale information and a physically accurate model that includes all relevant effects (baryonic physics, intrinsic alignments) to maximize the utility of full-sky data. To achieve this, I aim to execute flagship multi-GPU simulations and inference—e.g., a 2048^3 realization projected to `NSIDE`$= 1024$ (Górski et al., 2005; Zonca and others., 2019), reaching beyond $\ell \sim 3000$—on $\mathcal{O}(64\text{–}512)$ *H100* GPUs. I envision a state-of-the-art full-field inference package capable of analyzing real data from *Euclid* and *LSST* in a reasonable timeframe (a few weeks), supported by rigorous software development, physical modeling, and validation.

In parallel, I will continue developing a spatially aware component-separation framework for *primordial B-modes*, targeting controlled bias in the *tensor-to-scalar ratio* r. The end-to-end differentiable graph maps (r, foreground hyperparameters) to multi-frequency skies—including *beams*, *bandpasses*, *scanning*, *noise*, and *masking*—moving away from parametric methods to simulation-based inference to further reduce r-bias on real data from *SO*.

Being part of *SO* and *LSST DESC* and having worked on both sides of the cosmological timeline—early-time *CMB* and late-time *LSS*—allows me to leverage my experience and actively pursue cross-correlations between the two probes. Cross-correlations between CMB and LSS are powerful tools to mitigate systematics and tighten cosmological constraints. My goal is to use tSZ and temperature maps as a prior for LSS baryonic physics and use galaxy shear and density maps to assist with CMB component separation and delensing.

Beyond scientific goals, I will continue teaching and mentoring students and researchers in **JAX** and differentiable programming for cosmology, and more generally in modern high-performance computing techniques. As my scientific goals have significant overlap with the computational challenges faced by the community, I see great value in sharing knowledge to help others adopt these new tools.

Finally, I will continue to emphasize open, reusable outputs—maintained packages, benchmarks, and data artifacts—so results transfer directly to upcoming analyses in *Euclid*, *LSST DESC* and *SO*. I will continue supporting community adoption of **JAX** in cosmology and contribute selectively to performance-critical kernels where bespoke implementations materially advance *weak-lensing* and *CMB* inference.

3 Interest in the INFOCW project at IAP

Applying for this postdoctoral position within the INFOCW project led by F. Leclercq and G. Lavaux at IAP is a natural progression for my research. My work on differentiable simulations lies at the core of the project's methodology and aligns perfectly with the goals of the Aquila and "Learning the Universe" collaborations. The opportunity to collaborate closely with these groups offers an ideal environment to push forward modeling to the next level. I envision a tight feedback loop where my work on differentiable forward models is continuously refined by the advanced statistical methods and implicit (simulation-based) inference techniques developed within the team.

My technical contributions, particularly with `JaxPM` and `jaxdecomp`, are directly aligned with the project's mission to develop innovative forward modeling and Bayesian inference methods for Euclid. I am extremely passionate about the intersection of high-performance computing, software engineering, and cosmology, and I have made it a central pillar of my PhD research. I have led the development of critical **JAX**-native infrastructure for the community, including distributed FFTs and particle-mesh solvers, and I am eager to bring this expertise to IAP.

My objective is to drive the scientific exploitation of *Euclid* early data releases (DR1/DR2), specifically targeting galaxy clustering and weak lensing analysis. This is a multifaceted challenge requiring significant software engineering and validation; as a postdoc, I would not only lead the pipeline development but also mentor students and researchers to build a sustainable effort around these new techniques.

I am particularly eager to apply these pipelines to *Euclid* data, whose full-sky coverage provides the perfect playground for field-level inference. My work aligns closely with the Aquila Consortium's tradition of innovative, open astronomical software. The **JAX**-native components I have built—from CMB component separation (**FURAX**, `jax-healpy`) to distributed PM engines (`jaxdecomp`, `JaxPM`)—are engineered as community packages. At IAP, I will consolidate these into shared infrastructure with stable APIs, benchmarks, and tutorials, ensuring they serve as reproducible foundations for the community. Furthermore, I will continue to lead **JAX** pedagogy, accompanying the team as we transition to modern, distributed workflows that serve *Euclid* and the broader cosmological community.

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